

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459789.8243 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.052
Transit depth (Rp/Rs)^2	2.79 +/- 1.75 [%]
Orbital Inclination (inc)	85.77 +/- 1.58
Airmass coefficient 1 (a1)	1.22 +/- 0.61
Airmass coefficient 2 (a2)	-0.14 +/- 0.13
Scatter in the residuals of the lightcurve fit is	48.67 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	6.5
Transit Duration (day)	0.055 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.052 vs 0.1418 (deviation 0.39σ)
Duration fit vs published	0.055 d vs 0.1196 d (deviation 1.16σ)
Mid-time O-C	-6.9 min
Depth SNR	2.6
Residual scatter	48.67 %

Light curve

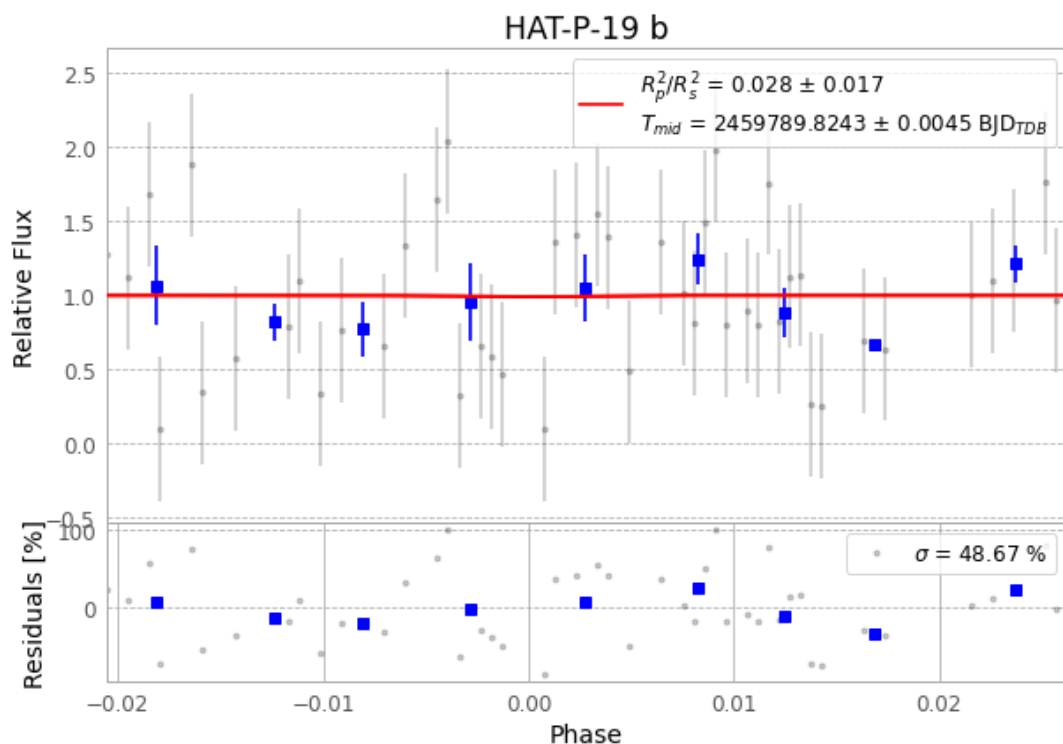


Figure 1 — FinalLightCurve_HAT-P-19 b_2022-07-28.png (SHA-256 66bbab94addc4e30...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7be1c31e4ec47e17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), HATP-19220729054513.FITS → HATP-19220729101813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220729.log (f559237d3b81...), FinalLightCurve_HAT-P-19 b_2022-07-28.png (66bbab94addc...), FinalLightCurve_HAT-P-19 b_2022-07-28.csv (7767e3be5b51...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:00Z.